Customer, Persona, Feature, and Business Benefit Note

Purpose and product framing

This note consolidates the customer and product material in the project folder, especially the MINEASK persona workflow analysis, the mining-sector persona profiles, and the GeoMine/MINERVA product and solution documents. The folder reflects multiple product stages and names—MINERVA, MINEASK, and GeoMine Intelligence—so this note treats them as related iterations of a mining intelligence proposition, not as proof that every described capability is already live in one release.

The shared proposition is: **help mining-sector decision makers assess concessions, assets, compliance, operational context, and risk by bringing fragmented mining, geospatial, regulatory, environmental, and market information into one analytical workspace, then producing evidence-backed answers and reports.**

The most consistently supported current value is rapid, structured **query-and-assess** work. Continuous monitoring, automated alerts, customized reporting, and some integrations are identified as roadmap or partial capabilities in the persona workflow analysis. The detailed agent claims there should therefore be verified against the deployed version before being used in a sales commitment.

Who the customers and clients are

“Customer” means the organization that buys or funds access. “Client” can also mean the end client served by a consultant, law firm, or technical adviser using the platform to deliver work.

Primary organizational customer groups

1. **Mining and exploration companies:** operators, concession holders, exploration companies, and project developers screening new sites, planning expansion, or monitoring existing assets.
2. **Mining consultants and technical advisers:** geology, mining, environmental, and investment advisers who need to answer client questions and produce defensible reports efficiently.
3. **Investors and lenders:** commercial banks, private equity and venture capital, royalty/streaming firms, and other project financiers assessing title, infrastructure, ESG, and operating exposure.
4. **Government and regulators:** Minerals Commission and other public bodies responsible for license oversight, inspection prioritization, land-use monitoring, and environmental compliance.
5. **Mine-support suppliers and contractors:** drilling, equipment, fuel, power, water, logistics, maintenance, security, catering, laboratory, and camp-service providers seeking demand and prospect intelligence.
6. **Insurance and risk organizations:** underwriters and risk teams evaluating physical, climate, geotechnical, logistics, and liability exposure.

Typical buying roles

Economic buyers are likely to include mine executives, country managers, heads of exploration or operations, consulting-firm partners, investment directors, bank credit heads, regulator program leads, and supplier business-development leaders. Functional users include analysts, geologists, GIS staff, ESG/compliance professionals, lawyers, mine planners, field teams, and risk/security analysts. Procurement authority and willingness to pay are not validated consistently in the folder; treat them as hypotheses to confirm with customer interviews and pilots.

Shared feature set and what it enables

Feature or capability	What it does	User-facing benefit
Site and concession selection	Start from a concession/permit identifier, coordinates, map pin, district, route, community, or uploaded boundary	Reduces friction in initiating an assessment and gives the analysis a defined geography

Feature or capability	What it does	User-facing benefit
Joined mining and geospatial data	Brings together cadastre/title, geology, terrain, water, roads, infrastructure, settlements, protected areas, satellite, weather, and market information	Replaces repeated manual searches across disconnected sources and makes evidence easier to compare
Site/concession scorecards and screening	Structures geological, legal, access, ESG, and operational factors into a consistent assessment	Supports prioritization and portfolio comparison; helps expose deal-breakers earlier
Map studio and spatial overlays	Visualizes boundaries, infrastructure, environmental and community context, hazards, and routes	Makes location-specific risk and opportunity understandable to technical and executive users
AI-assisted questions, agents, and reports	Produces a structured dossier or answer from a query; persona workflow material describes specialist agents for cadastre, EITI, satellite, environment, logistics, and valuation	Accelerates first-pass analysis and report preparation; user still needs to validate source data and judgment
Regulatory rules and compliance context	Maps licensing steps and regulatory requirements; some material describes rules based on Ghana's Minerals and Mining Act 703	Helps surface likely process steps, missing items, or legal hard stops; does not replace counsel or official confirmation
ESG, land, and community exposure analysis	Screens water stress, vegetation/forest change, protected areas, tailings, settlements, community incidents, and land context	Helps teams identify and investigate environmental/social issues before decisions or fieldwork
Logistics and infrastructure analysis	Evaluates roads, haulage corridors, ports, power, water, processing access, and route context	Supports access planning, feasibility review, route risk, and infrastructure dependency analysis

Feature or capability	What it does	User-facing benefit
Mine-support demand intelligence	Identifies mining clusters, project stages, service gaps, potential clients, and outreach priorities	Helps suppliers allocate sales effort and build a more qualified prospect pipeline
Collaboration, provenance, and exports	Shared projects, comments, version history, source lineage/confidence, dashboards, and exports are described across platform documentation	Supports review, auditability, team handoffs, and client-ready or committee-ready communication
Monitoring and alerts (roadmap/partial)	Scheduled re-runs, change detection, and alerting are described as planned or not yet solved for several personas	Could shift value from one-off assessments to ongoing risk monitoring once implemented and validated

Persona-by-persona value and financial relevance

Financial relevance below describes the mechanism by which a persona could affect cash flow, cost, capital allocation, or loss exposure. The folder does not provide verified realized ROI for these personas. Savings and avoided losses should be measured in pilots rather than represented as guaranteed outcomes.

1. Mining / minerals consultant

Organization / client: Independent mining, geological, environmental, or transaction adviser; the consultant's paying client may be an investor, concession applicant, operator, or lender.

Job and pain: Turn a location or concession question into a defensible recommendation. Today this can involve manually collecting title, royalty, regulatory, road, protected-area, conflict, land, and satellite information, then assembling a report.

Most relevant features: Location/concession lookup; parallel data gathering; cadastre and fiscal-history checks; licensing pathway; logistics and land/community brief;

satellite/ASM risk screening; source-aware dossier and field verification checklist; report export/white-labeling (some reporting features remain partial or planned).

Benefits: Faster first-pass due diligence; repeatable methodology across client engagements; easier identification of issues requiring field or legal verification; more time for interpretation and client advice.

Financial/business relevance: The clearest direct value case is lower analyst hours and cost per engagement, more engagements per consultant, faster client turnaround, and improved margin on fixed-fee work. It can also reduce professional exposure by showing evidence and open verification items. The persona analysis claims a manual week of record-pulling can become an afternoon of structured analysis; this is a product-document estimate, not independently validated. Track baseline and post-pilot hours per report, revisions, turnaround, and gross margin.

2. ESG and compliance manager

Organization / client: Mining operator, project developer, contractor, investor, or a consulting firm supporting environmental and social compliance.

Job and pain: Screen environmental and social exposure, maintain permit awareness, and report against relevant frameworks and regulator requirements.

Most relevant features: Protected-area buffers; water stress/watershed context; forest and vegetation change; tailings proximity; community and incident context; permit-condition screening; map-backed evidence and framework-specific reporting (some framework reporting is only partly addressed); scheduled monitoring is a roadmap item in the workflow analysis.

Benefits: Earlier visibility into environmental and social constraints; more efficient screening across sites; better preparation for permitting, field inspection, stakeholder engagement, and reporting; clearer documentation of what is known versus what needs verification.

Financial/business relevance: Potentially reduces duplicated consultant/data-gathering costs, late redesign or mitigation costs, delays caused by missed constraints, and expected costs from non-compliance, remediation, or stakeholder conflict. For an operator, the financial value is risk avoidance and smoother project progression; for an adviser, it is reduced assessment effort and faster deliverables. Quantify with hours to compile a screening pack, number of issues found before fieldwork, and schedule/cost

changes attributable to early discovery. The product is a screening aid, not a guarantee of compliance.

3. Mining investment analyst / lender / investment director

Organization / client: Bank, private equity or venture fund, royalty/streaming company, project financier, or corporate development/investment team.

Job and pain: Decide whether an asset merits investment or financing by checking title, regulatory standing, infrastructure, environmental exposure, commodity context, and comparable projects; make a defensible recommendation.

Most relevant features: Concession/title verification; overlap and dispute review; license status and regulatory hard stops; infrastructure and ESG risk mapping; commodity-price and peer context where available; standardized scorecards; decision-ready investment/credit memo; portfolio comparison. The workflow notes economic modeling and go/no-go recommendations are partly supported and that the tool supplements rather than replaces financial models. Post-investment monitoring is roadmap/not yet.

Benefits: Faster screening of more opportunities; earlier discovery of deal-breakers; more consistent diligence across assets; improved ability to direct expensive technical, legal, and field diligence toward higher-potential targets.

Financial/business relevance: Supports capital allocation and may reduce diligence cost per opportunity, wasted diligence on unsuitable assets, and expected loss from title, infrastructure, ESG, or regulatory surprises. The largest financial leverage is better selection and risk pricing on large capital decisions, but no investment return can be attributed to the platform alone. Measure screening cycle time, diligence spend before/after, share of opportunities stopped early for documented reasons, and post-close forecast-versus-actual risk findings.

4. Legal and compliance manager

Organization / client: Mining operator, concession applicant, lender/investor, law firm, or corporate legal/compliance function.

Job and pain: Verify whether rights are valid, identify overlaps and regulatory traps, advise on licensing steps, and create an auditable legal position.

Most relevant features: Cadastre holder/boundary/status and expiry checks; overlap/dispute markers; imagery-based boundary comparison; royalty/payment history where available; versioned and reviewable regulatory rules; evidence lineage and confidence; explicit checklist for matters requiring registry or counsel verification.

Benefits: Faster preliminary title and licensing review; fewer manual lookups; more consistent issue spotting; reviewable regulatory logic and clearer audit trail. The platform's own persona analysis is explicit that some encumbrances and formal evidence requirements remain partly addressed and must be verified with official registries.

Financial/business relevance: Can reduce routine legal research hours and avoid wasted transaction or application spend where a fatal defect is identified early. Better process visibility can reduce delay and legal rework. The financial downside avoided may include sunk diligence costs, disputes, penalties, or license loss, but these are contingent risks, not guaranteed savings. Measure lawyer hours per preliminary review, issues escalated before commitment, and time to produce the supporting evidence pack.

5. Mine operations manager / planning team

Organization / client: Operating mine, mine developer, or contractor responsible for planning, infrastructure, and site operations.

Job and pain: Understand site conditions and nearby constraints affecting pits, waste dumps, processing, haul roads, water, utilities, communities, and risk zones. Some MINEASK material also describes telemetry and periodic operational indicators; the broader product documents emphasize planning and spatial intelligence.

Most relevant features: Internal mine-plan/data inputs; spatial overlays; terrain, hydrology, infrastructure, and settlement context; route analysis; change detection and encroachment/risk layers; mine planning exports to GIS/CAD and mining software. The persona workflow explicitly says MINEASK is not a live dispatch/control room; real-time operations control and continuous automated monitoring are gaps or roadmap items.

Benefits: Better-informed planning and route/site choices; reduced time reconciling internal and external spatial data; earlier awareness of physical, community, environmental, or security constraints; improved coordination between planning, ESG, and operations.

Financial/business relevance: Potential contribution to lower haulage time/fuel, fewer planning changes, reduced equipment or infrastructure exposure, and less disruption

from overlooked constraints. Benefits depend on data freshness and integration into operating decisions. Do not claim live productivity optimization or dispatch savings unless deployed functionality supports it. Measure planning hours, route distances/cycle times where analyzed, rework, and disruption events associated with mapped risks.

6. Risk and security manager

Organization / client: Mining operator, insurer, logistics provider, lender, or corporate risk function.

Job and pain: Understand incidents, illegal mining, transport corridor threats, unrest, political risk, and exposure around sites and concessions, then communicate where action or further investigation is needed.

Most relevant features: Threat-radius and threshold filters; incident/event data; ASM encroachment and satellite layers; corridor scoring; political/unrest overlays; concession intersection; risk dossier. The workflow says emerging-conflict early warning is partial and 24/7 alerting/continuous monitoring is not yet available.

Benefits: Faster periodic risk assessment; better prioritization of sites and routes for review; shared evidence for security planning and leadership decisions; reduced dependence on scattered incident sources.

Financial/business relevance: Potentially reduces expected costs from theft, route interruption, injury, asset damage, and delayed shipments by improving prioritization and contingency planning. It may also support more informed insurance and logistics decisions. It is not a substitute for live security operations or verified intelligence. Measure analyst hours per site/route assessment, incidents with prior mapped indicators, route disruption duration, and security spend targeted to higher-risk areas.

7. Mine-support supplier / contractor business-development manager

Organization / client: Drilling, equipment, fuel, logistics, power, water, maintenance, laboratory, security, catering, or other mining supplier.

Job and pain: Identify active and upcoming mines, expansions, exploration clusters, and regions where supplier coverage is thin; decide which prospects sales teams should pursue.

Most relevant features: Mine/project and concession discovery; project-stage and activity context; service-gap and distance-to-demand analysis; prospect lists and outreach prioritization; map-based regional activity views.

Benefits: More focused prospecting; shorter time spent building lead lists; stronger territory planning; better match between service offering and mine/project need.

Financial/business relevance: Direct commercial value comes from lowering customer-acquisition cost and sales research time, increasing qualified meetings and win rate, and directing sales resources toward regions with demand. The supplier could also avoid underutilized inventory or equipment deployment in low-demand areas. Track cost per qualified lead, conversion rate, sales-cycle length, revenue per territory, and utilization. Lead accuracy and procurement data coverage are critical dependencies.

8. Government regulator / public-sector oversight team

Organization / client: Minerals Commission, environmental agencies, district/public authorities, or other public institutions overseeing licenses, land use, environment, and mining activity.

Job and pain: Monitor concession status and overlap, prioritize inspections, identify disturbance or illegal mining signals, and prepare evidence for enforcement and policy.

Most relevant features: Regulatory dashboards; concession/land-use intersections; satellite disturbance/change mapping; inspection-priority lists; environmental/community overlays; district activity summaries and evidence packages.

Benefits: Better geographic visibility; more targeted inspection planning; less manual collation; improved consistency in monitoring and reporting; support for public transparency and evidence-led oversight.

Financial/business relevance: Enables more efficient use of inspection budgets and staff time, potentially increases coverage per officer, and may reduce public remediation and revenue-leakage exposure when issues are identified sooner. These benefits are public value and fiscal efficiency rather than direct platform revenue for the regulator. Measure inspections prioritized per analyst hour, field visits that confirm actionable issues, response time, and reporting effort. Public data rights, agency workflows, procurement, and trust are adoption dependencies.

9. Credit risk officer / insurance underwriter

Organization / client: Commercial or investment bank, insurer, reinsurer, or underwriting syndicate.

Job and pain: Assess borrower or insured-asset validity, physical hazards, climate/water exposure, logistics interruption, ESG liabilities, and ongoing compliance risk.

Most relevant features: Concession audits; infrastructure dependency maps; flood/slope/water/climate context; route and bridge exposure; satellite disturbance monitoring; ESG/community assessment; credit/underwriting reports and (planned) alerts.

Benefits: More consistent risk intake; clearer asset-level evidence; better questions for technical diligence; improved risk segmentation and monitoring where repeat analysis is available.

Financial/business relevance: May reduce underwriting and credit-review cost, improve pricing/terms based on better risk differentiation, and reduce expected loss from unrecognized hazards or interruption exposure. This is decision support and should not replace engineering surveys, actuarial models, or credit judgment. Measure review time, external diligence spend, risk findings discovered before binding/closing, and loss/claim experience over time.

Cross-persona benefit summary

Persona	Main value lever	Financial/business outcome to measure
Consultant	Faster client analysis and report production	Hours per engagement, turnaround, engagements per team, margin
ESG/compliance	Earlier risk identification and less reporting duplication	Screening cost, early issues found, rework/delay, remediation exposure
Investment analyst/lender	Better diligence prioritization and capital allocation	Diligence cost, time-to-decision, bad opportunities stopped early, risk surprises

Persona	Main value lever	Financial/business outcome to measure
Legal/compliance	Faster preliminary verification and traceable evidence	Legal research hours, early defects found, transaction/application rework
Mine operations/planning	Better spatial and infrastructure decisions	Planning time, route/fuel metrics, rework, disruption and downtime exposure
Risk/security	Better prioritization of sites and corridors	Assessment time, incident response/disruption cost, security resource allocation
Supplier/contractor BD	More efficient prospecting and territory targeting	Qualified leads, acquisition cost, conversion, sales cycle, utilization
Government regulator	More targeted inspections and oversight	Cost per actionable inspection, coverage, response time, staff reporting time
Credit/insurance	More consistent asset-level risk review	Review time, diligence cost, risk pricing quality, losses/claims over time

Practical sales and product implications

- Lead with the most concrete current outcome: **faster, more structured, evidence-backed assessment of a defined site, concession, route, or portfolio question**.
- Tailor the output to the buyer: consultant dossier; investor/credit memo; legal evidence and verification checklist; ESG screening pack; operations map; supplier prospect list; regulator inspection queue.
- Sell time savings and decision quality as measurable pilot hypotheses, not guaranteed ROI. Agree baseline measures before a pilot.
- Make data provenance, date, geographic coverage, confidence, and “must verify” items visible. This is especially important for legal, investment, regulatory, and insurance users.

- Separate current capabilities from roadmap items in proposals. The persona analysis flags continuous monitoring, automated alerts, some framework reporting, white-label exports, and live operational control as incomplete or future capability.
- Prioritize customer discovery with consultants and legal/due-diligence users for assessment workflows, and suppliers for prospect intelligence. Validate budget owner, procurement path, data trust, and willingness to pay before relying on assumed subscription economics.

Source files consulted

- `Marketing/MINEASK_Persona_Workflow_Analysis.html` — six persona workflows, current/partial/roadmap coverage, and explicit scope boundaries.
- `Marketing/Mining_Sector_Persona_Profiles.html` — broader role and task profiles across mining-sector professionals.
- `Week 2 - MVP Build/Aino.ai Mining Docs/1. codex/md/00-overview/user-personas.md` — executive, exploration, operations, ESG, supplier, and regulator needs.
- `Week 2 - MVP Build/Aino.ai Mining Docs/1. codex/md/00-overview/product-vision.md` and `01-homepage/platform-capabilities.md` — product proposition and shared features.
- `Week 2 - MVP Build/Aino.ai Mining Docs/1. codex/md/06-solutions/` — customer segments and use cases for mining companies, exploration teams, regulators, investors/lenders, and support services.
- `Initial Research/3. artifacts/product_strategy_writeup.md` — caveats around demand validation and unsupported pricing assumptions.